

Basic Networking and Firewalls

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Strategic Knowledge and Inclusive Lifelong Learning in the health sector for youth Employability and Development

- 1 Networking Foundations
- 2 Hospital Network Architecture
- 3 Firewall Fundamentals
- 4 Packet Filtering
- 5 Stateful and Proxy Firewalls
- 6 Deployment Patterns
- 7 VPNs and Secure Remote Access

- 8 Hands-On Lab: Firewall Simulation
- 9 Optional: Cloud Computing
- 10 Final Integration

Learning path

We start from the language of networks: **hosts, addresses, packets, ports, and protocols**. Then we move to **hospital network zones, firewalls, DMZs, VPNs**, a **hands-on firewall lab**, and finally an optional view of **cloud computing**.

Hospital storyline

St. Isidore Hospital operates an EHR, laboratory system, pharmacy system, imaging archive, patient portal, guest Wi-Fi, medical devices, remote access for clinicians, and some cloud services. Networking decides which systems can reach each other; firewalls enforce that decision.

Where this fits

Networking is the map on which many security controls operate. Firewalls control traffic paths, authentication protects remote entry points, encryption protects communication, and access control limits what authenticated users can do after they reach an application.

- **Encryption:** TLS, VPNs, and IPsec protect traffic across untrusted networks.
- **Access control:** network segmentation reduces which systems can even be contacted.
- **Authentication:** VPNs, proxies, cloud consoles, and administrative interfaces need strong identity checks.
- **Privacy and compliance:** network security supports confidentiality, integrity, availability, and incident containment.

Networking Foundations

Section Context: Networking Foundations

What we are talking about

A network lets systems exchange data. Security begins by understanding what communicates, through which path, using which protocol, and for what operational purpose.

Hospital lens

When a ward workstation opens an EHR page, it may use DNS, IP routing, TCP, TLS, HTTP, database connections, authentication services, and logging systems. A firewall rule sees only part of that story unless the architecture is documented.

Hosts, Clients, and Servers

Host

A host is any device with a network identity: workstation, server, printer, phone, medical device, container, virtual machine, or firewall interface.

Client and server

A client initiates a request. A server waits for requests on a network address and port. The same machine can be a client in one flow and a server in another.

Hospital example

A lab analyzer is a client when it sends results to the laboratory system. The laboratory system is a server for the analyzer, but a client when it forwards verified results to the EHR.

IP Addresses

Addressing

IP gives hosts logical addresses so packets can be routed across interconnected networks [Postel, 1981]. IPv4 addresses look like 192.168.10.42; IPv6 addresses are longer and written in hexadecimal groups.

Hospital example

The EHR application server may live at 10.20.5.15. A ward workstation at 10.30.2.44 can reach it only if routing and firewall policy allow the path.

Private and Public Addresses

Private addresses

Hospitals usually use private address ranges internally, such as 10.0.0.0/8, 172.16.0.0/12, or 192.168.0.0/16. These are not directly routed on the public Internet.

Public addresses

Internet-facing services need public reachability, often through a firewall, load balancer, reverse proxy, or cloud edge service.

Subnets



Core idea

A subnet groups addresses that belong to the same local network segment. CIDR notation describes the network size: 10.30.2.0/24 means addresses from 10.30.2.0 to 10.30.2.255.

Hospital example

St. Isidore may assign 10.10.0.0/24 to servers, 10.20.0.0/24 to clinical workstations, 10.30.0.0/24 to medical devices, and 10.90.0.0/24 to guest Wi-Fi.

Ports and Services

Port

A port identifies a service endpoint on a host. The address tells us which host; the port tells us which application or service on that host.

Hospital example

The same server might expose HTTPS on port 443, SSH administration on port 22, and a database listener on port 5432. A good firewall policy does not treat these as the same risk.

Common Ports

Port	Protocol	Typical use
53	DNS	Name resolution
80 / 443	HTTP / HTTPS	Web applications and APIs
25 / 587	SMTP	Email relay and submission
22	SSH	Remote administration
3389	RDP	Windows remote desktop
5432 / 3306	PostgreSQL / MySQL	Database access

Security point

Opening a port means exposing a service. The risk depends on the service, who can reach it, and how it is authenticated, patched, and monitored.

Packets and Flows

Packet

A packet carries headers and payload. Headers contain delivery information such as source, destination, protocol, and ports. Payload contains the data being carried.

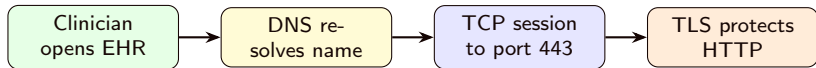
Flow

A flow is the conversation formed by related packets, such as a workstation connecting to the EHR over HTTPS.

Hospital example

A firewall may log a flow as: source 10.20.0.45, destination 10.10.0.12, protocol TCP, port 443, action allowed.

Scheme: From User Action to Network Flow



Reading the scheme

A simple click in the browser becomes multiple network events. Security controls may operate at different points in that path.

TCP and UDP

TCP

TCP is connection-oriented. It establishes a session, tracks sequence numbers, retransmits lost data, and is common for web, email, remote administration, and many databases [Eddy, 2022].

UDP

UDP is connectionless. It has less overhead and is common for DNS, some real-time traffic, and protocols where the application handles reliability [Postel, 1980].

DNS, DHCP, NAT

DNS

DNS translates names into addresses. If DNS fails or is poisoned, users may not reach the correct service.

DHCP

DHCP assigns network configuration automatically, such as IP address, gateway, and DNS server.

NAT

Network Address Translation maps internal addresses to other addresses, often allowing many private hosts to share public Internet connectivity.

Recap: Networking Foundations

Takeaways

- Networks are built from hosts, addresses, ports, protocols, packets, and flows.
- Security decisions should be based on documented communication needs.
- A network path is not automatically safe just because the application login works.

In St. Isidore Hospital

The hospital must know which systems talk to which services before it can write meaningful firewall rules or detect suspicious traffic.

Hospital Network Architecture

Section Context: Hospital Network Architecture



What we are talking about

Enterprise networks evolved from centralized systems to LANs, interconnected internal networks, WANs, Internet connectivity, remote access, and cloud services. Each step increases reachability and the need for controlled boundaries.

Hospital lens

St. Isidore no longer has one mainframe and terminals. It has clinical workstations, servers, medical devices, mobile devices, partner connections, public services, and cloud platforms.

LAN, WAN, and Internet

LAN

A Local Area Network connects systems in a limited area, such as a hospital building, ward, or data center.

WAN

A Wide Area Network connects sites across larger distances, such as a main hospital, rural clinic, and administrative office.

Internet

Internet connectivity is operationally essential but exposes internal resources to external threats unless carefully mediated.

Network Zones

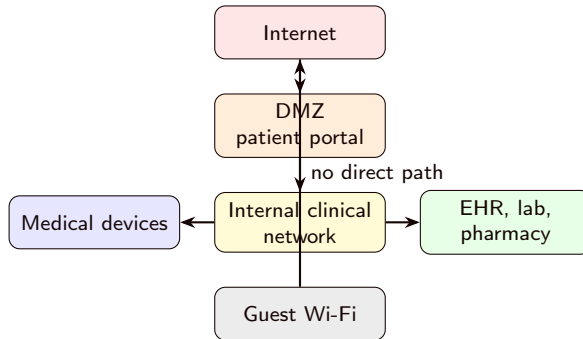
Zone

A zone is a group of systems with similar trust level, exposure, and security requirements.

Hospital example

Useful zones include public services, DMZ, clinical workstations, servers, medical devices, guest Wi-Fi, administration, backup, and security monitoring.

Scheme: Hospital Zones



Design point

Zones reduce blast radius. A compromised guest phone should not be able to scan medication cabinets or database servers.

Segmentation

Core idea

Segmentation separates systems so that compromise in one area does not automatically imply access to all other areas.

Hospital example

Imaging devices may need to send studies to PACS, but they should not browse the payroll database, reach backup servers, or initiate arbitrary outbound Internet connections.

Choke Points

Firewall placement

A firewall should create a controlled point where traffic can be allowed, denied, logged, and investigated. The traditional goal is that all traffic between zones passes through the firewall.

Hospital example

If the patient portal can talk directly to the EHR database without crossing a monitored control point, the hospital loses visibility and containment.

Recap: Hospital Network Architecture

Takeaways

- Hospital networks should be described as zones and flows, not only as devices.
- Segmentation supports safety, privacy, resilience, and incident containment.
- Choke points matter because controls and logs need a place to operate.

In St. Isidore Hospital

St. Isidore should separate public services, clinical systems, devices, guest access, and administration before deciding detailed firewall rules.

Firewall Fundamentals

Section Context: Firewall Fundamentals

What we are talking about

A firewall protects a host or network by controlling network traffic according to security policy. It is not a complete security system, but it is a central enforcement and monitoring point.

Hospital lens

The hospital firewall does not know whether a diagnosis is correct, but it can prevent the guest Wi-Fi network from reaching the EHR database port.

Firewall Goals

Classic design goals

- All relevant traffic should pass through the firewall.
- Only traffic authorized by policy should pass.
- The firewall itself should be hardened and resistant to compromise.

Hospital example

Internet users may reach the public patient portal, but not internal laboratory analyzers or the database backend.

Firewall Policy

Policy question

Which traffic should be allowed, between which sources and destinations, using which protocols, ports, users, applications, time windows, and logging requirements?

Hospital example

Allow clinical workstations to reach the EHR on HTTPS. Allow the EHR application server to reach the database. Deny direct workstation-to-database access.

Positive and Negative Filtering

Positive filtering

Allow only traffic that matches approved criteria. This is conservative: default deny, then add what the hospital needs.

Negative filtering

Block traffic that matches forbidden criteria. This is easier at first, but it assumes administrators know every dangerous pattern in advance.

Default Action

Default deny

Everything not explicitly allowed is blocked. This is usually preferred for sensitive healthcare networks.

Default allow

Everything not explicitly blocked is allowed. This may reduce support tickets, but it often leaves unexpected paths open.

Firewall Limitations

A firewall cannot

- protect against traffic that bypasses it;
- stop every insider threat;
- inspect traffic it cannot see or decrypt;
- compensate for unpatched exposed services;
- protect a laptop infected outside the hospital before it reconnects internally.

Recap: Firewall Fundamentals

Takeaways

- Firewalls enforce network policy at controlled points.
- Default deny is safer for clinical and administrative systems.
- Firewalls reduce risk, but do not replace patching, authentication, endpoint security, and monitoring.

In St. Isidore Hospital

The hospital should use firewalls to make expected traffic explicit and unexpected traffic visible.

Packet Filtering

Section Context: Packet Filtering

What we are talking about

Packet filtering firewalls decide whether to forward or discard packets based on information in network and transport headers.

Hospital lens

A packet filter can allow 10.20.0.0/24 to reach the EHR web server on TCP port 443, while denying guest Wi-Fi access to the same destination.

What Packet Filters Inspect

Typical fields

- source and destination IP address;
- protocol, such as TCP, UDP, or ICMP;
- source and destination port;
- input or output interface;
- selected TCP flags, such as ACK.

Rule Order Matters

Top-down evaluation

Firewall rules are usually evaluated from top to bottom. The first matching rule decides the action. The final rule should make the default action explicit.

Hospital example

If a broad “allow clinical network to any database port” rule appears before a narrow deny rule, the narrow deny rule may never be reached.

Example Rule Set

#	Source	Destination	Service	Action
1	Clinical LAN	EHR web	TCP 443	Allow
2	EHR web	EHR DB	TCP 5432	Allow
3	Clinical LAN	EHR DB	TCP 5432	Deny
4	Guest Wi-Fi	Internal nets	Any	Deny
5	Any	Any	Any	Deny

Reading the table

Users reach the application, not the database. The application reaches the database. Guest Wi-Fi is kept out of internal networks.

SMTP Example: Why Details Matter

Problem

Simple SMTP rules may allow inbound traffic to high-numbered client ports or allow any internal host to send mail directly to the Internet.

Hospital example

If every workstation can send SMTP to the Internet, malware on one workstation can send spam or exfiltrate data. A safer policy forces outbound mail through the hospital mail gateway.

Packet Filtering Strengths

Advantages

- simple and fast;
- transparent to users;
- suitable for hardware implementation;
- useful for broad segmentation and baseline policy.

Packet Filtering Weaknesses

Limitations

- it does not understand application functions;
- logging is limited to inspected fields unless configured carefully;
- advanced user authentication is difficult at packet level;
- TCP/IP quirks and fragmentation can create bypass risks;
- misconfiguration is a major security flaw.

Attacks on Packet Filters

IP spoofing

An attacker sends packets with a forged internal source address.

Countermeasure: drop packets with internal source addresses arriving from external interfaces.

Source routing

An attacker tries to specify the packet route to bypass policy.

Countermeasure: block source-routing options.

Tiny fragments

An attacker fragments packets so the first fragment hides important TCP header fields. Countermeasure: require enough header information before accepting fragments.

Recap: Packet Filtering

Takeaways

- Packet filters are fast and useful, but they see limited context.
- Rule order and default action are security-critical.
- Header tricks, broad rules, and weak logging can undermine the design.

In St. Isidore Hospital

St. Isidore should use packet filtering for clear zone boundaries, but not rely on it to understand clinical application behavior.

Stateful and Proxy Firewalls

Section Context: Stateful and Proxy Firewalls

What we are talking about

Traditional packet filters inspect each packet in isolation. Stateful and proxy firewalls add more context: connection state, transport sessions, application behavior, or user identity.

Hospital lens

The firewall should know whether an inbound packet is part of a legitimate existing connection from the EHR server, or an unsolicited attempt from the Internet.

Stateful Packet Inspection

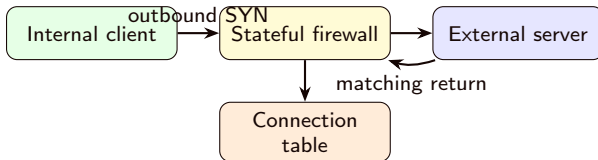
Core idea

A stateful firewall records active connections. It allows return traffic only when it matches a known established flow.

Hospital example

If a pharmacy workstation opens an HTTPS session to an update service, return packets for that session are allowed. Unsolicited inbound packets to random high ports are denied.

Scheme: Stateful Firewall



Application-Level Gateway

Application proxy

An application-level gateway understands a specific application protocol. It can authenticate users, inspect application commands, enforce protocol-specific policy, and produce richer logs.

Hospital example

A web proxy can allow staff to browse approved medical resources while blocking known malicious sites, suspicious downloads, or unauthorized upload destinations.

Application Proxy Trade-Off

Benefit

The proxy can make more meaningful decisions than a packet filter because it sees application-level semantics.

Cost

The proxy adds overhead and complexity. It splits communication into two connections: client to proxy, and proxy to remote service.

Circuit-Level Gateway

Core idea

A circuit-level gateway also creates two TCP connections, but it focuses on transport-level session control rather than inspecting application content.

Hospital example

A SOCKS gateway can decide whether a workstation may establish an outbound connection, without understanding the full application protocol running through that connection.

Comparing Firewall Types

Type	Sees	Good for
Packet filter	IPs, ports, protocol, interface	Fast baseline filtering
Stateful firewall	Packet fields plus connection state	Safer return traffic handling
Circuit gateway	Transport-level sessions	Controlled outbound connectivity
Application proxy	Application protocol and sometimes user identity	Deep policy and better logs

Recap: Stateful and Proxy Firewalls

Takeaways

- Stateful firewalls reduce risks created by dynamic client ports.
- Application proxies can enforce richer policy, but add overhead.
- Firewall type should match the policy question being asked.

In St. Isidore Hospital

St. Isidore may use stateful firewalls between zones and application proxies for Internet browsing, mail, or exposed web services.

Deployment Patterns

Section Context: Deployment Patterns

What we are talking about

Firewall value depends on placement. The same rule behaves differently at the Internet edge, between internal zones, on a server host, or inside a cloud security group.

Hospital lens

An edge firewall protects from the Internet, but it may not stop a compromised ward workstation from reaching a database unless internal segmentation also exists.

Bastion Host

Definition

A bastion host is a hardened system identified as a critical security point. It often hosts an application proxy, circuit gateway, jump service, or administrative entry point.

Hardening

Install only essential services, restrict privileges, protect configuration, keep executable code read-only where possible, log access carefully, and monitor changes.

Host-Based and Personal Firewalls

Host-based firewall

A software firewall protects one server or workstation. It can apply rules tailored to that host's role, independent of network topology.

Personal firewall

A lighter endpoint firewall blocks unexpected inbound access and may monitor outbound application activity to detect malware or unsafe software.

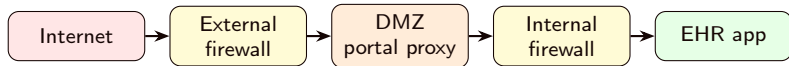
Definition

A demilitarized zone is a network segment for systems that must be reachable from outside but should not sit directly inside the trusted internal network.

Hospital example

The patient portal reverse proxy belongs in the DMZ. The EHR database does not.

Scheme: DMZ with Internal Firewall



Reading the scheme

The DMZ is not trusted like the internal network. If a DMZ host is compromised, the internal firewall still limits what the attacker can reach next.

Distributed Firewalling

Core idea

Modern networks combine edge firewalls, internal firewalls, host-based rules, cloud security groups, and endpoint controls under one administrative policy.

Hospital example

Even if a workstation is inside the hospital building, its local firewall can still block unexpected incoming connections from another compromised workstation.

Monitoring and Logs

Why logs matter

Firewall logs help reconstruct attempted connections, denied traffic, unusual flows, scans, policy violations, and possible compromise.

Operational need

Logs should be aggregated with alerts and other telemetry. Isolated firewall logs are useful; linked firewall, identity, endpoint, and application logs are much more useful.

Recap: Deployment Patterns

Takeaways

- Placement determines what a firewall can see and enforce.
- DMZs protect public-facing services without placing them fully inside the trusted network.
- Distributed firewalling and log aggregation are necessary in modern environments.

In St. Isidore Hospital

The hospital should not rely only on one Internet-edge firewall. Internal segmentation and host-based controls reduce lateral movement.

VPNs and Secure Remote Access

Section Context: VPNs and Secure Remote Access

What we are talking about

A VPN uses cryptography and authentication to create a protected connection across an untrusted network. It is useful for remote sites and remote users, but it also creates an entry point into the organization.

Hospital lens

Doctors on call, external specialists, remote clinics, and administrators may need access from outside the hospital. The VPN must protect traffic and control who enters.

What a VPN Provides

Confidentiality

Traffic is encrypted so outsiders cannot easily read it.

Authentication

The VPN endpoint authenticates users, devices, sites, or gateways before allowing traffic.

Integrity

Cryptographic checks help detect modification of protected traffic.

IPsec Scenario

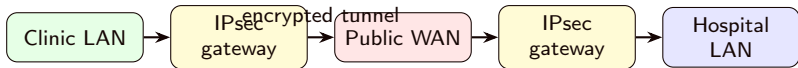
Site-to-site VPN

Two hospital sites can use IPsec gateways to protect traffic across a public WAN. Internal traffic may be ordinary IP inside each LAN, while traffic crossing the public network is encrypted.

Hospital example

The rural clinic sends laboratory orders to the main hospital. IPsec protects the inter-site path, but internal firewalls still decide which applications the clinic can reach.

Scheme: Site-to-Site VPN



VPN Placement

Inspection problem

If traffic is encrypted before it reaches an internal firewall, the firewall may not be able to inspect application details. VPN placement affects visibility and control.

Hospital example

If a remote user's VPN tunnel drops them directly into the clinical network, the hospital may have less segmentation than expected. A better design terminates VPN access in a controlled zone.

VPN Is Not Full Trust

Common mistake

Authenticating to the VPN should not mean unrestricted internal access.

Hospital example

An external radiologist may need PACS and imaging reports, not payroll, pharmacy stock systems, or domain controller administration.

Recap: VPNs and Secure Remote Access

Takeaways

- VPNs protect traffic over untrusted networks using cryptography and authentication.
- VPN placement affects firewall visibility.
- Remote access should still be segmented and least-privilege.

In St. Isidore Hospital

St. Isidore should treat VPN as a secure path to controlled services, not as a magic door into the whole hospital network.

Hands-On Lab

Section Context: Firewall Simulation Lab

What we are talking about

Before using a real firewall, it helps to practice with a controlled model. The lab uses Python to simulate packets, zones, firewall rules, stateful replies, default-deny behavior, and firewall logs.

Hospital lens

St. Isidore wants to check whether doctors can reach the EHR, whether guest Wi-Fi is isolated, whether the EHR application can reach its database, and whether the database blocks direct access from workstations.

Lab Tool Choice

Notebook simulator

The notebook in `lesson-06/examples` is a **teaching firewall**. It does not protect the computer running it. It makes the decision process visible: each packet is matched against ordered rules, and every decision is logged.

Realistic next steps

Mininet can emulate hosts, switches, links, and Linux network stacks on one machine, while **Scapy** can craft and inspect packets. They are powerful, but they usually require root or a Linux VM, so they fit better after the concepts are clear.

Lab 1: Packet Filtering

Goal

Model the hospital network as zones and write rules that describe expected traffic.

- allow clinical workstations to reach the EHR web service on TCP 443;
- allow only the EHR application server to reach the EHR database on TCP 5432;
- block guest Wi-Fi from internal services;
- log the first rule that matched each packet.

Lab 2: Rule Order and Default Deny

Goal

See why firewall policy is not just a list of wishes. It is an ordered decision procedure.

Hospital example

If a broad allow rule is placed before a deny rule, the deny rule may never run. The lab changes the order and shows how the same packet receives a different decision.

Lab 3: Stateful Replies

Goal

Compare a stateless filter with a stateful firewall.

Hospital example

A doctor opens an HTTPS connection to the EHR. The request goes from the clinical LAN to the EHR server, but the response comes back in the opposite direction. A stateful firewall can allow the reply because it belongs to an established flow.

Lab 4: Reading Firewall Logs

Goal

Use logs to explain policy behavior instead of guessing.

- identify which rule allowed or denied each connection;
- count denied attempts by source zone;
- find policy mistakes, such as direct database access from workstations;
- connect log evidence to segmentation decisions.

Recap: Hands-On Lab: Firewall Simulation

Takeaways

- A firewall policy is an executable form of the network design.
- Rule order, default action, and statefulness change real outcomes.
- Logs turn firewall behavior into evidence that can be reviewed and improved.

In St. Isidore Hospital

The lab turns St. Isidore's network policy into something testable: expected clinical flows work, unexpected paths are blocked, and the reason is visible.

Optional: Cloud Computing

Section Context: Optional Cloud Computing

What we are talking about

Cloud computing moves some infrastructure, platforms, or applications outside direct hospital ownership. The same security goals remain, but responsibilities are shared with the provider.

Hospital lens

St. Isidore may use a cloud email platform, cloud backup, cloud analytics, or a hosted patient portal. Network security now includes provider APIs, identity, encryption, logs, regions, and contracts.

NIST Cloud Definition

Cloud model

NIST defines cloud computing as on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort [Mell and Grance, 2011].

Five essential characteristics

On-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.

Service Models

SaaS

The provider runs the application. The consumer uses it through a web or application interface.

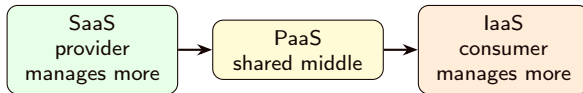
PaaS

The provider runs the platform. The consumer deploys applications onto that platform.

IaaS

The provider supplies infrastructure such as virtual machines, storage, and networks. The consumer controls more of the system stack.

Scheme: Shared Responsibility



Security point

The hospital never outsources accountability completely. It must understand which controls are run by the provider and which remain the hospital's responsibility.

Deployment Models

Public cloud

Cloud infrastructure is offered to many customers by a provider.

Private cloud

Cloud infrastructure is used exclusively by one organization.

Community and hybrid cloud

Community cloud is shared by organizations with common requirements. Hybrid cloud combines multiple cloud or on-premises environments with portability or integration.

Cloud Actors



NIST reference architecture

NIST describes five main actors: cloud consumer, cloud provider, cloud auditor, cloud broker, and cloud carrier [Liu et al., 2011].

Hospital example

St. Isidore is the consumer. The cloud platform is the provider. A security assessor may act as auditor. A managed-service partner may broker or aggregate services. Network carriers provide connectivity.

Cloud Security Threats

Typical risks

Cloud introduces or amplifies risks such as abuse of cloud resources, insecure APIs, malicious insiders, shared technology weaknesses, data loss, account hijacking, and unclear risk visibility.

Hospital example

A stolen cloud administrator account can expose backups, analytics datasets, network rules, or identity settings. MFA and logging are not optional decorations; they are core controls.

Cloud Data Protection

Data states

Health data must be protected in transit, at rest, and during use. Encryption helps, but key management determines who can actually read the data.

Trade-off

Encrypting data before sending it to the cloud can reduce provider-insider risk, but may limit server-side computation unless special techniques are used.

Security as a Service

Idea

Cloud providers and security providers can offer identity management, data loss prevention, web and email security, security assessments, intrusion management, SIEM, encryption, and continuity services.

Hospital example

St. Isidore might use cloud email security to filter phishing, a SIEM to aggregate firewall and identity logs, and cloud backup for disaster recovery.

Recap: Optional: Cloud Computing

Takeaways

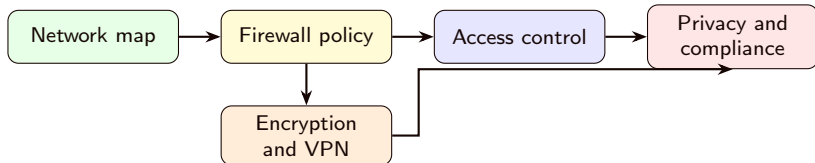
- Cloud is still networking, identity, access control, encryption, logging, and contracts.
- Service model determines who controls which security layer.
- Healthcare cloud adoption requires explicit responsibility, auditability, and key-management decisions.

In St. Isidore Hospital

St. Isidore can use cloud safely only if it understands provider responsibility, hospital responsibility, data location, identity controls, and monitoring.

Final Integration

Final Concept Map



Reading the map

The network map identifies possible paths. Firewall policy controls those paths. Access control limits actions after reaching a system. Encryption protects traffic. Privacy and compliance requirements shape how personal and health data must be protected in practice.

Final Recap



Core lesson

Network security is not only blocking the Internet. It is designing and monitoring the expected communication paths between hospital systems.

Healthcare security lesson

A well-designed hospital network makes clinical work possible while limiting unnecessary reachability, containing compromise, preserving logs, and protecting patient data.

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